

Week 3 Class Notes

Preference Measurement

TEMBA (Section II)
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Consolidated from Week 3 Pre-Class Notes + in-class discussion.

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Measuring & Quantifying Consumer Preferences

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EXAM NOTE Quiz 2 moved to Session 4, rescoped to CBC interpretation only – see Topic 6. The MaxDiff/TURF/ratings-conjoint material in this chapter is still core course content.

Measuring & Quantifying Consumer Preferences

Conjoint = Consider + Jointly

1. Big-Picture Summary

KEY Two tools, one agenda: **MaxDiff** answers “what matters most” (a ranking); **Conjoint Analysis** answers “how much does it matter, in dollars” (willingness-to-pay).

Both tools exist because direct questions fail – stated preferences (“what do you want?”) and stated importance ratings (“rate this 1–9”) both cost the respondent nothing, so people either ask for everything or rate everything as important. Forcing a real trade-off and observing the choice is what reveals true value.

EXAM Quiz 2 is **not** in this session – it was moved to Session 4 (Sep 22) and rescoped to CBC interpretation only (total utility, attribute importance, WTP, incremental contribution). The MaxDiff/TURF/ratings-conjoint material below is still core course content, just not this particular quiz’s material – see Week 4 Pre-Class Notes for the official Quiz 2 scope.

MAXDIFF & TURF ANSWER...

“What matters most, and in what mix?”

A ranking of items by revealed best/worst choice (MaxDiff), then the bundle that reaches the most unique people (TURF) – not stated importance.

CONJOINT ANALYSIS ANSWERS...

“How much, in dollars?”

Willingness-to-pay for every attribute level, estimated from the trade-offs respondents actually make between priced bundles.

Topics 2–6 build each tool from its formula up, in its own color, with its own worked numbers (a GPS device, a self-driving car, a credit-card perk bundle) so the mechanics stop being abstract.

What’s due / where things stand

ITEM	STATUS
Capstone Milestone 2	Due the night before Session 3
Quiz 2	Moved to Session 4 (Sep 22); CBC interpretation only – see Week 4 Pre-Class Notes
This session’s material	MaxDiff, TURF, ratings-based conjoint, CBC fundamentals – core content, exam-fair game generally

Topic 2. MaxDiff (Best-Worst Scaling)

Respondents see a small set of items (4–5 typical) and pick the ONE they like most and the ONE they like least – items in the middle stay unranked for that question; repetition across questions, with different item pairings, is what produces a full ranking.

Best use case: 8–30 features/benefits/ideas where you need to know which ones matter most, without rating-scale bias.

Net choice score = Most% – Least%, counted over EVERY appearance of that item – not just the times it was actually selected

COMMON MISTAKE The denominator trap: if an item appeared 3 times and was picked Most twice, the rate is $2 \div 3 = 66.7\%$ – not $2 \div 2$. “Neither selected” still counts as an appearance.

Design math

Item slots = (number of items) $\times$ (appearances per item). Tasks per respondent = item slots $\div$ (items shown per task).

DESIGN CHOICE

WHY IT MATTERS

Vary item pairings and on-screen position

Software-balanced designs avoid confounding position with preference

Keep each task to 4–5 items

More items per task raises cognitive load without adding information

COMMON MISTAKE Vague/overlapping item wording

“Free banking” vs. “no annual fee”, or bundling two distinct items as one, muddies results



Memory aid

“Most minus Least, over every appearance” – never divide by selections only.

Worked Example — Pecan Street Bank: Choosing the Perks to Score

MAXDIFF

Problem setup: Pecan Street Bank is choosing which four of twelve possible credit-card perks to offer. 200 respondents, each perk appearing 3 times (600 total appearances), scored via MaxDiff net choice score – the bundle decision itself (TURF, under a \$60/year budget) continues in Topic 3.

Solution — MaxDiff scoring

Groceries: 302 Most ($302/600 = 50.3\%$), 39 Least ($39/600 = 6.5\%$) → net score = $50.3\% - 6.5\% = +43.8\text{pp}$

Resulting ranking (best to worst net score): groceries (+43.8pp), no annual fee (+31.2pp), gas/EV (+29.3pp), fraud alerts + lock (+20.5pp), phone protection (−3.3pp), 0% intro APR (−7.8pp), no foreign fees (−8.0pp), purchase protection (−9.2pp), streaming credit (−10.5pp), travel insurance (−21.5pp), late-fee forgiveness (−23.5pp), lounge access (−41.0pp).

Key terms

TERM	SIMPLE EXPLANATION	WHY IT MATTERS	EXAMPLE/ANALOGY
MaxDiff / Best-Worst Scaling	Repeatedly pick the most- and least-important item from a small set	Forces a real trade-off, unlike a 1–9 importance rating	Choosing your favorite and least-favorite of 4 card perks, several times over
Net choice score	Most% − Least%, over every appearance	Ranks items by revealed preference, not stated importance	Groceries: $50.3\% - 6.5\% = +43.8\text{pp}$

Topic 3. TURF (Total Unduplicated Reach & Frequency)

KEY INSIGHT The highest-scoring individual items don't automatically make the best bundle – if two high scorers are liked by the same people, adding the second adds little new reach. TURF picks the combination that reaches the most *unique* people, not the combination with the highest average score.

TURF DOES	BEST USE CASE
Portfolio optimization	% of people who find at least one item in a group appealing
Redundancy reduction	Avoids picking items that appeal to the exact same people
Bundle selection	Deciding which product bundles, flavors, or marketing messages to offer together

Memory aid

“Total Unduplicated Reach and Frequency” – unduplicated is the whole point: a second popular item only helps if it reaches *new* people.

Worked Example — Pecan Street Bank: Picking the Bundle Under Budget

TURF

Continuing Topic 2's Pecan Street Bank case: MaxDiff scored and ranked all 12 perks by net choice score. Now TURF narrows that ranking to the 4-perk bundle that reaches the most people under a \$60/year cost cap.

Solution — TURF bundle selection under budget

Picking the top 4 items by net score alone (groceries, no annual fee, gas/EV, fraud alerts) reaches only 49% of customers (98 of 200) – too much overlap between fans of similar perks.

BUNDLE	FOUR PERKS	COST/YEAR	REACH
A	Groceries, gas/EV, intro APR, foreign fees	\$70	87%
B	Groceries, intro APR, foreign fees, fraud alerts	\$59	80%
C	Groceries, gas/EV, foreign fees, late-fee forgiveness	\$54	70.5%
D	Groceries, gas/EV, foreign fees, travel insurance	\$60	69.5%

Interpretation

Under a \$60 budget, Bundle A no longer qualifies (over budget). Among the feasible bundles, **Bundle B gives the highest reach (80%) at \$59** – the answer isn't “best overall bundle,” it's “best bundle that fits the constraint.”

Marginal reach walkthrough for Bundle B (160 of 200 people, 80%): Groceries 69 new → 69 reached. + No foreign fees 48 new → 117. + 0% intro APR 41 new → 158. + Fraud alerts 2 new → 160 (80%).

EXAM Only *new* people count toward reach at each step – someone already reached by an earlier item doesn't add again when a later item also covers them. This is why the last perk added only 2 new people: almost everyone who liked it already liked something earlier in the bundle.

Key terms

TERM	SIMPLE EXPLANATION	WHY IT MATTERS	EXAMPLE/ANALOGY
TURF	Total Unduplicated Reach and Frequency – picks the bundle covering the most unique people	A high-scoring item can be redundant if it reaches people already covered	Choosing 4 card perks to maximize % of customers satisfied by at least one
Reach	% of people reached by at least one item in a bundle	The TURF optimization target	Bundle B reaches 80% of 200 surveyed customers

Topic 4. The MaxDiff → TURF Pipeline

A single pipeline, run in sequence:

STEP 1 — MAXDIFF	STEP 2 — CONVERSION	STEP 3 — TURF
Run a MaxDiff study to score and rank the full item list	Convert MaxDiff utility/preference scores into binary data (e.g. top choices or threshold passes)	Run TURF on that binary data to find the optimal mix – e.g. which 3 flavors or 4 features cover the widest market



Memory aid

Score first, convert to binary second, optimize the bundle third – never skip the conversion step.

SEE ALSO This pipeline run end to end, on real numbers: Topic 2 (MaxDiff scoring) and Topic 3 (TURF bundle selection) walk the same Pecan Street Bank credit-card-perks case through all three steps above.

Topic 5. Conjoint Analysis • Fundamentals

A product is a bundle of attributes (e.g. a credit card is brand × interest rate × annual fee × rewards × credit limit). Learn how buyers value each component and you can design the product that maximizes profitability, not just guess.

KEY INSIGHT The conjoint principle: don't ask people what they want – observe what they choose when they can't have it all. The pattern of choices reveals the implicit value (**partworth**) of each attribute level.

$$\text{The additive utility model: } U_{\text{profile}} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

Each x_i is a dummy variable marking whether a profile has a given attribute level; each β_i – the partworth – is that level's contribution to total utility.

The four-step workflow

STEP 1	STEP 2	STEP 3	STEP 4
Vary product features to build product profiles	Ask respondents to rate / rank / choose among profiles	Use regression to estimate part-worth utilities	Compute WTP, importance, and simulate markets

Dummy coding: the N–1 rule

An attribute with N levels needs only N–1 dummy columns – the omitted level (the **reference level**) gets a partworth of 0 by definition; every other level's partworth is measured relative to it.

Worked illustration from class: for 3 mutually exclusive levels (A, B, C), build only 2 columns – one for A, one for B. If a profile has A=0 and B=0, it must be C by elimination, so C never needs its own column. Anything not explicitly included is implicitly 0.

COLUMN A	COLUMN B	IMPLIED LEVEL
1	0	A
0	1	B
0	0	C (the reference level)

Memory aids

- **Dummy coding:** “N levels, N–1 columns – the one you leave out is your reference.”
- **Partworth:** the “worth” of one **part** (attribute level) of the whole product, in utils.
- **\$/util converter:** always built from the *price* attribute – “this much price partworth = this much real money,” then multiply any other partworth by that same \$X to get its WTP.
- **Importance vs. WTP:** importance is a *relative* (within-this-study, sums to 100%) measure; WTP is an *absolute* dollar measure. An attribute can be “low-importance but high-dollar” – e.g. self-driving: 7% importance but ≈ \$5,900.

Designing a conjoint study

DESIGN CHOICES	COMMON PITFALLS
4–8 attributes, 2–5 levels each	Too many attributes → respondent fatigue
Always include price, for WTP	Unrealistic combinations → disengagement
Use images for realism; pre-test	Ignoring interactions → biased estimates
Sample: <100 for ratings, 200–500+ for CBC	No “None” option in CBC → overstated demand

Worked Example 1 — Handheld GPS Device

RATINGS-BASED CONJOINT • OLS

Problem setup: four binary attributes – Accuracy (5ft vs. 50ft), Battery (32hr vs. 12hr), Display (3D vs. 2D), Price (\$150 vs. \$250). $2^4 = 16$ possible profiles; full factorial design shows the respondent all 16, each rated 1–100.

$$\text{Rating} = \beta_0 + \beta_1(\text{Accuracy}) + \beta_2(\text{Battery}) + \beta_3(\text{Display}) + \beta_4(\text{Price}) + \varepsilon$$

Solution

VARIABLE	PARTWORTH (β)	T-STAT
Intercept	2.7	2.7
5ft vs. 50ft (Accuracy)	9.6	10.9
32hr vs. 12hr (Battery)	30.4	34.5
3D vs. 2D (Display)	14.9	16.9
\$150 vs. \$250 (Price)	40.6	46.1

$R^2 = 0.99$, all coefficients highly significant.

Interpretation — converting partworths to dollars (WTP)

$$\text{Value of 1 util} = \text{Price} \div \text{Price partworth} = \$100 \div 40.6 \approx \$2.46/\text{util}$$

FEATURE UPGRADE	PARTWORTH	× \$2.46/UTIL	WTP
Accuracy: 50ft → 5ft	9.6	×\$2.46	\$23.65
Battery: 12hr → 32hr	30.4	×\$2.46	\$74.88
Display: 2D → 3D	14.9	×\$2.46	\$36.70

KEY INSIGHT Decision rule: include a feature only if its WTP exceeds the cost to include it, by a comfortable margin. Industry example: Marriott designed Courtyard from scratch using conjoint – business travelers valued a large desk and good lighting far more than a fancy lobby, becoming one of the most successful hotel launches in history.

Worked Example 2 — The Value of Self-Driving

RATINGS-BASED CONJOINT • FRACTIONAL FACTORIAL

EXAM – INSTRUCTOR’S OWN WORKED EXERCISE

Problem setup: attributes are Brand (Tesla/BYD/Hyundai), Price (\$30K/\$45K/\$60K), Package (1 Basic, 2 Mid, 3 Luxury), Drive Mode (Regular, Self-Driving). $3 \times 3 \times 3 \times 2 = 54$ combinations – infeasible to show all, so a fractional factorial design selects 13 profiles; each respondent rates all 13 on a 1–100 scale.

Reference levels here: Package 1, \$30K, Regular drive, Tesla – each gets partworth 0 by the N–1 rule above.

Solution

COEFFICIENT	VALUE	P-VALUE
Intercept	36.02	0.006
Package 2	51.62	0.002
Package 3	69.02	0.001
\$45,000	–27.75	0.014
\$60,000	–60.34	0.005
Self-Drive	11.85	0.120
BYD	–6.89	0.512
Hyundai	–20.40	0.067

COMMON MISTAKE Self-Drive’s partworth (11.85) has $p = 0.12$ – only marginally significant with just 13 observations from one respondent. A small sample can make a real effect look statistically weak.

Importance weights (3 steps)

1. Adjust partworths so the minimum within each attribute = 0.
2. Range = max adjusted partworth – min adjusted partworth, per attribute.
3. Importance = range ÷ sum of all ranges.

ATTRIBUTE	RANGE (UTILS)	IMPORTANCE
Package	69.02	43%
Price	60.34	37%
Brand	20.40	13%
Drive Mode	11.85	7%
Total	161.62	100%

Package/amenity level matters most (43%), then price (37%); self-driving carries the smallest relative importance (7%) – importance and WTP-in-dollars can tell different stories (a low-importance attribute can still carry real dollar value).

Dollar value of self-driving

Step 1 – build the converter: $PW(\$30K) - PW(\$60K) = 0 - (-60.34) = 60.34 \text{ utils} = \$30,000$, so 1 util $\approx$ \$497.

Step 2 – $WTP(\text{Self-Drive}) = 11.85 \text{ utils} \times \$497/\text{util} \approx \$5,892$.

Managerial read: this consumer would pay roughly \$5,900 more for self-driving – profitable to include if it costs less than that to add. Tesla's brand value over Hyundai $= 20.40 \times \$497 \approx \$10,139$.

Practice questions EXAM

Same partworths and \$497/util converter

QUESTION	SOLUTION
Q1. Brand value of Tesla relative to BYD?	$PW(\text{Tesla}) - PW(\text{BYD}) = 0 - (-6.89) = 6.89 \text{ utils}$; $WTP = 6.89 \times \$497 \approx \$3,424$
Q2. Value of the Luxury package (Pkg 3) over Basic (Pkg 1)?	$PW(\text{Pkg3}) - PW(\text{Pkg1}) = 69.02 - 0 = 69.02 \text{ utils}$; $WTP = 69.02 \times \$497 \approx \$34,303$
Q3. Predicted rating for a Hyundai, \$45K, Package 2, Self-Driving profile?	$36.02 + 51.62 + (-27.75) + 11.85 + (-20.40) = 51.34$
<i>Method for Q3:</i> start from the intercept, then add the coefficient for every dummy that's "on" in the profile (Package 2, \$45,000, Self-Drive, Hyundai) – skip any dummy for an attribute level not present (Tesla/BYD dummies stay off since this profile is Hyundai; the Package 3 dummy stays off since this profile is Package 2).	

WTP decision framework

$WTP = \text{price ceiling}$. $\text{Cost} = \text{price floor}$. Include a feature only if $WTP > \text{Cost} + \text{margin}$.

KEY INSIGHT WTP is the maximum a consumer would pay – the market price actually charged depends on competition, heterogeneity, and strategy on top of that ceiling. $WTP \neq \text{market price}$: a monopolist can capture most of the surplus between cost and average WTP; a competitive market pushes price back toward cost.

COMMON MISTAKE Average WTP can hide real heterogeneity – some customers might pay \$15K for self-driving, others \$0. Conjoint reveals the full demand curve, not just its average.

Key terms

TERM	SIMPLE EXPLANATION	WHY IT MATTERS	EXAMPLE/ANALOGY
Dummy coding (N–1 rule)	An attribute with N levels needs only N–1 dummy columns	The omitted (reference) level's partworth is 0 by definition	3 mutually exclusive levels A/B/C need only 2 columns (for A and B)
Partworth (β)	The utility contributed by one attribute level	The fundamental output of any conjoint model	Battery 32hr vs. 12hr is worth 30.4 utils to this consumer
Additive utility model	$U = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon$	Lets regression decompose overall preference into per-attribute value	Total GPS rating = sum of accuracy + battery + display + price partworths
Reference (omitted) level	The one level per attribute with no dummy variable; its partworth = 0	Every other level's partworth is measured relative to it	Tesla, \$30K, Package 1, Regular drive in the self-driving example
Willingness-to-pay (WTP)	Partworth $\times$ (\$ per util)	Converts abstract utils into a real decision number	Self-driving is worth $\approx$ \$5,892 to this consumer
Importance weight	An attribute's range $\div$ sum of all ranges	Shows which attribute swings choices most – relative, not dollar, measure	Package = 43% importance vs. Drive Mode = 7%

Topic 6. Choice-Based Conjoint (CBC)

Choice-based conjoint is what's used in the real world over ratings-based conjoint.

RATINGS-BASED CONJOINT	CHOICE-BASED CONJOINT (CBC)
Rates each profile on a scale	Chooses among 3–5 profiles, can include “None”
Analyzed with OLS regression	Mimics real purchase decisions
Simple to implement	Analyzed with logit models – industry standard today

Why CBC dominates in practice: (1) ecological validity – choosing is what consumers actually do; (2) lower cognitive burden than rating many profiles; (3) a “None” option captures non-buyers; (4) software handles complex designs automatically.

$$\Pr(\text{choose } j) = \frac{e^{V_j}}{\sum_k e^{V_k}}, \quad \text{where } V_j = \beta_1 x_{j1} + \beta_2 x_{j2} + \cdots + \beta_K x_{jK}$$

Estimated by maximum likelihood or Bayesian methods – not OLS. Higher utility → higher choice probability, but never certainty; produces market-share predictions naturally.

Hierarchical Bayes (HB) can estimate individual-level partworths from choice data – a preference “ruler” per respondent, not one average customer.

KEY INSIGHT Fractional factorial / orthogonal designs select a subset of profiles so every pair of attribute levels appears together equally often – unbiased main-effect estimates with far fewer profiles. The key assumption: no interaction effects; when interactions matter, larger designs are needed.

Incentive alignment: telling respondents 1-in-N will actually receive a real product based on their conjoint answers produces more reliable data than a purely hypothetical task (a GPS study gave 1-in-100 an actual device; an auto study gave the winner a shot at a real \$40K car).

Memory aid

“Choose, don’t rate” – CBC mimics an actual purchase decision and is analyzed with a logit model (MLE/Bayesian), not OLS.

Tool comparison

TOOL	STRENGTHS	BEST FOR
Sawtooth	Gold standard; HB built in	Professional studies
Qualtrics	Survey platform with conjoint module	Quick studies, CRM integration
Excel + OLS	Simple, transparent	Classroom exercises
R / Python	Full control, free	Academic research

For class projects: start with ratings-based conjoint + Excel regression. For consulting or major decisions: use Sawtooth CBC/HB.

Key terms

TERM	SIMPLE EXPLANATION	WHY IT MATTERS	EXAMPLE/ANALOGY
Choice-Based Conjoint (CBC)	Respondent chooses among 3–5 full profiles (often with a “None” option)	Mimics a real purchase decision; today’s industry standard	“Which of these 3 phone plans would you buy – or none?”
Logit model	$\Pr(\text{choose } j) = e^{V_j} \div \sum e^{V_k}$	Converts utility differences into choice probabilities for CBC data	Higher utility → higher probability, never certainty
Hierarchical Bayes (HB)	Estimates individual-level partworths from choice data	A preference “ruler” per respondent, not one average customer	Two respondents can get different personal WTP-for-self-driving estimates
Orthogonal / fractional factorial design	A subset of profiles where every pair of attribute levels appears together equally often	Makes main effects estimable without showing every possible combination	13 profiles instead of all 54 in the self-driving exercise
Incentive alignment	Make the survey choice have a real consequence (a chance to win the chosen product)	Produces more reliable, carefully-considered answers	1-in-100 respondents actually received the GPS device they helped design

Topic 7. Practice Drill

Same format as the Quiz 1/2 voice-prep drills: work each question before reading its solution. New scenario below, not the chapter's own worked numbers – so this actually tests recall. One question per topic above, interleaved – not grouped by tool – since that's closer to how exam questions actually come at you.

QuickCharge power bank, CBC study. Attributes: Capacity (10,000mAh vs. 20,000mAh), Price (\$25 vs. \$40), Charging Speed (Standard vs. Fast). Reference levels: 10,000mAh, \$25, Standard. Partworths: Capacity 20,000mAh = +0.50; Price \$40 = -0.90; Charging Speed Fast = +0.70.

Q1 – MaxDiff, concept

A respondent sees an item in 4 MaxDiff screens and picks it as “Most” once and never as “Least.” What's its net choice score?

- A. $1 \div 1 = 100\%$ B. $1 \div 4 - 0 \div 4 = 25\%$ C. $1 \div 4 = 25\%$, ignoring Least entirely D. Cannot be computed from one appearance

Correct: B. Most% – Least% = $1/4 - 0/4 = 25\%$, using every appearance (4) as the denominator for both, not just the time it was actually picked (A's mistake) or Most alone (C's).

Q2 – TURF, concept

Two perks each have the highest individual MaxDiff scores in their category, but 90% of people who like perk X also already like perk Y. What does TURF say about adding X to a bundle that already has Y?

- A. Always add it – it has a high score B. It adds little reach, since most of its fans are already reached by Y C. It doubles the bundle's reach D. TURF ignores individual scores entirely

Correct: B. A high score doesn't guarantee new reach – TURF optimizes for unique people covered, and X's fans overlapping 90% with Y's means X would mostly reach people the bundle already has.

Q3 – Total utility

Using the QuickCharge table above: total utility of a 20,000mAh, \$40, Fast-charging profile?

- A. $0.50 + 0.90 + 0.70 = 2.10$ B. $0.50 - 0.90 + 0.70 = 0.30$ C. $0.50 - 0.90 - 0.70 = -1.10$ D. $0.90 - 0.50 - 0.70 = -0.30$

Correct: B. Sum one partworth per attribute, keeping each sign as given: $0.50 + (-0.90) + 0.70 = 0.30$. A's mistake is dropping the price partworth's negative sign – always include price even when (especially when) it subtracts.

Q4 – Importance weight

Ranges: Capacity 0.50, Price 0.90, Charging Speed 0.70. What's Price's importance?

- A. 90% B. $0.90 \div (0.50 + 0.90 + 0.70) = 42.9\%$ C. $0.90 \div 0.50 = 180\%$ D. 33%, since there are 3 attributes

Correct: B. Importance = that attribute's range $\div$ the sum of every attribute's range = $0.90 \div 2.10 \approx 42.9\%$ – D's mistake (splitting evenly across attribute count) ignores that the attributes don't swing choices equally.

Q5 – WTP

The \$40 level costs \$15 more than the \$25 reference and its partworth is -0.90 . What's the \$/util rate, and the WTP for the Fast-charging upgrade (partworth $+0.70$)?

- A. Rate = $0.90/\$15 = 0.06$ utils/\$; WTP = $0.70 \div 0.06 \approx \$11.67$ B. Rate = $\$15/0.90 \approx \$16.67/\text{util}$; WTP = $0.70 \times 16.67 \approx \11.67 (same answer, built the other way) C. WTP = $0.70 \times \$15 = \10.50 D. WTP = $\$15 - 0.70 = \14.30

Correct: A or B – same number, $\approx \$11.67$. Build the converter from price first (always), then apply it to the upgrade's own utility gain. C and D skip building a real \$/util rate and just combine the raw numbers, which isn't a unit-consistent calculation.

Q6 – CBC vs. ratings, concept

QuickCharge's team wants to know what % of the market would actually buy at each price point, not just which profile scores highest on average. Which design should they run, and why?

- A. Ratings-based, OLS – simplest to run B. CBC with a "None" option, analyzed with a logit model – mimics real purchase decisions and produces market-share-style predictions C. Either works identically for this question D. MaxDiff alone, since it ranks fastest

Correct: B. A ratings scale (A) never asks anyone to actually choose or decline, so it can't estimate real purchase/no-purchase rates; CBC's "None" option and logit-based choice probabilities are built for exactly this question. MaxDiff (D) ranks importance, not purchase likelihood.